

Chapter 1: Differential Calculus & Optimization

1.1 The Definition of the Derivative

Instantaneous Rate of Change as a Limit

Differential calculus studies rates of change and slopes of curves. The derivative of a function $f(x)$ at point x represents the slope of the tangent line to the graph at that point, defined formally as the limit of the difference quotient as h approaches zero.

Key Formula / Law: Limit Definition of Derivative: $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

1.2 Essential Rules of Differentiation

Power, Product, Quotient, and Chain Rules

- Power Rule: $\frac{d}{dx} [x^n] = n \cdot x^{(n-1)}$ - Product Rule: $\frac{d}{dx} [u(x) \cdot v(x)] = u'(x) \cdot v(x) + u(x) \cdot v'(x)$ - Quotient Rule: $\frac{d}{dx} \left[\frac{u(x)}{v(x)} \right] = \frac{u'(x)v(x) - u(x)v'(x)}{[v(x)]^2}$ - Chain Rule for composite functions: $\frac{d}{dx} [f(g(x))] = f'(g(x)) \cdot g'(x)$

1.3 Critical Points, Extrema, and Applied Optimization

First and Second Derivative Tests

A critical point occurs where $f'(c) = 0$ or $f'(c)$ is undefined. - First Derivative Test: If $f'(x)$ changes sign from positive to negative at c , $f(c)$ is a local maximum; if negative to positive, a local minimum. - Second Derivative Test: If $f'(c) = 0$ and $f''(c) > 0$, the curve is concave up and $f(c)$ is a local minimum. If $f''(c) < 0$, it is concave down and $f(c)$ is a local maximum.

In applied optimization, we formulate an objective function subject to physical or economic constraints, express it in terms of a single variable, and find its global extremum on the feasible domain.

Key Takeaways:

- Derivative $f'(x)$ indicates slope, instantaneous velocity, and direction of increase.
- Second derivative $f''(x)$ reveals concavity and inflection points.
- Always check boundary endpoints in constrained optimization problems.